Learning Objectives

- To learn about the air and water cooling techniques involved in two stroke and four stroke engines.
- To learn about various cooling components like cooling fan, cooling water, radiator and water jacket.

7.0 AIM

In internal combustion engines due to the combustion of the fuel air mixture, enormous heat is liberated inside the combustion chamber. From the total heat released, about 30-35% of the heat is transferred as useful mechanical work at the engine's crank shaft. Approximately 30% of heat is carried out by the burned gases. About 25- 30% of the heat is rejected by the cooling medium and the remaining is considered as the unaccounted loss which is due to the friction, radiation etc., The temperature of the combustion products immediately after the combustion of the fuel reaches the value which is even more than 2000°C. Due to the very high temperature the engine components are subjected to very high thermal stress and get expanded due to the more heat of the products. This leads to the burning of engine lubricating oil and more carbon will get deposited inside the combustion chamber. In order to avoid the damage of engine components and burning of lubricating oil the cooling system is essential which helps in sending the heat to the surroundings.

The following are effects of overheating of the engine components.

- Damage of the piston and piston sticking on the cylinder wall.
- Lubricating oil burning and formation of carbon deposits and on the combustion chamber parts and the valves.
- Burning of the engine valves (mainly the exhaust valve)
- Occurrence of Pre ignition, knocking and detonation
- Reduction in viscosity of the engine lubricating oil
- Wear and tear of the engine components
- More fuel consumption.

The following are effects of over cooling of the engine components.

- Power loss
- More fuel consumption
- Reduced Thermal efficiency
- Increase in viscosity of the engine lubricating oil
- Reduced mechanical efficiency

Characteristics of the efficient cooling system:

- From the overall heat produced, the cooling system must reject sufficient amount of heat about 28-30% from the overall heat produced from combustion of the fuel.
- Cooling system should be designed in such a way that the heat to be rejected quickly when the engine is operated at very high temperature.

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